

Assignment 5

Due date: Apr 24th, 2026 before midnight

Problem 1. Consider the following discrete time Markov process $\{X_n; n \geq 0\}$ with the unit interval $[0, 1]$ as state space. Its transformation law is given by

$$X_{n+1} = \begin{cases} \frac{1}{3} + \frac{2}{3}X_n, & \text{with prob. } X_n \\ \frac{2}{3}X_n, & \text{with prob. } 1 - X_n \end{cases}$$

Show that this process is a martingale.

Solution Compute the conditional expectation, for $n \geq 0$,

$$\mathbb{E}(X_{n+1} \mid X_n, \dots, X_0) = \left(\frac{1}{3} + \frac{2}{3}X_n\right)X_n + \frac{2}{3}X_n(1 - X_n) = X_n$$

Therefore X_n is a martingale.

Problem 2. Let $\{X_n, n = 1, 2, \dots\}$ be a martingale. Prove that for any set of positive integers $k \leq l < m$ that the difference $X_m - X_l$ is uncorrelated with X_k , that is,

$$\mathbb{E}[(X_m - X_l)X_k] = 0$$

Solution. Let $\{\mathcal{F}_n; n \geq 1\}$ be the filtration associated with the process $\{X_n, n = 1, 2, \dots\}$. Then,

$$\begin{aligned} \mathbb{E}[(X_m - X_l)X_k] &= \mathbb{E}[\mathbb{E}(X_k(X_m - X_l) \mid \mathcal{F}_k)] \\ &= \mathbb{E}[X_k \mathbb{E}((X_m - X_l) \mid \mathcal{F}_k)] \\ &= \mathbb{E}[\mathbb{E}(X_k \mid \mathcal{F}_k) - \mathbb{E}(X_l \mid \mathcal{F}_k)] \\ &= \mathbb{E}[X_k(X_k - X_k)] \\ &= 0, \end{aligned}$$

where we have used the martingale property $\mathbb{E}(X_m \mid \mathcal{F}_k) = X_k$ for any $m \geq k$ in the fourth equality.

Note: If you are not familiar with filtration, then you could also write $\{X_n, n = 1, 2, \dots\}$ is martingale with respect to some process $\{Y_1, Y_2, \dots\}$, and use the same method by conditioning on the process $\{Y_1, Y_2, \dots, Y_k\}$.

Problem 3. Assume that $Y_1, Y_2, \dots$ are independent and identically distributed with $\mathbb{P}(Y_1 = 1) = p$, $\mathbb{P}(Y_1 = -1) = q = 1 - p$ and $p \neq q$. Fix positive integers a and b . With $S_0 = 0$, and $S_n = Y_1 + \dots + Y_n$, $n \geq 1$, let

$$T = \min\{n : S_n = -a \text{ or } S_n = b\}.$$

- (a) Verify that $M_n = S_n - n(p - q)$ and $M_n = (q/p)^{S_n}$ are martingales.
- (b) Find the probabilities $\mathbb{P}(S_T = -a)$ and $\mathbb{P}(S_T = b)$.
- (c) Find $\mathbb{E}[T]$.

Solution.

- (a) To distinguish these two processes, let $X_n = S_n - n(p - q)$ and $Z_n = (q/p)^{S_n}$. Let $\{\mathcal{F}_n; n \geq 0\}$ be the filtration associated with the process $\{S_n, n \geq 0\}$. Note that $\mathcal{F}_n$ contains information about $\{S_0, S_1, \dots, S_n\}$.

We have

$$\begin{aligned}
\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) &= \mathbb{E}(S_{n+1} - (n+1)(2p-1) \mid \mathcal{F}_n) \\
&= \mathbb{E}(S_{n+1} \mid \mathcal{F}_n) - (n+1)(2p-1) \\
&= \mathbb{E}(S_n + Y_{n+1} \mid \mathcal{F}_n) - (n+1)(2p-1) \\
&= \mathbb{E}(S_n \mid \mathcal{F}_n) + \mathbb{E}(Y_{n+1} \mid \mathcal{F}_n) - (n+1)(2p-1) \\
&= S_n + \mathbb{E}(Y_{n+1}) - (n+1)(2p-1) \\
&= S_n + 2p - 1 - (n+1)(2p-1) \\
&= S_n - n(2p-1) = X_n
\end{aligned}$$

Thus $\{X_n; n = 1, 2, \dots\}$ is a martingale. We have

$$\begin{aligned}
\mathbb{E}(Z_{n+1} \mid \mathcal{F}_n) &= \mathbb{E}\left(Z_n \left(\frac{q}{p}\right)^{Y_{n+1}} \mid \mathcal{F}_n\right) \\
&= Z_n \mathbb{E}\left(\left(\frac{q}{p}\right)^{Y_{n+1}} \mid \mathcal{F}_n\right) \\
&= Z_n \mathbb{E}\left(\left(\frac{q}{p}\right)^{Y_{n+1}}\right) \\
&= Z_n \cdot \left(p \left(\frac{q}{p}\right)^1 + q \left(\frac{q}{p}\right)^{-1}\right) \\
&= Z_n \cdot (q + p) = Z_n
\end{aligned}$$

Hence, $\{Z_n; n = 0, 1, 2, \dots\}$ is a martingale.

(b) Apply martingale stopping theorem on process $Z_n = (q/p)^{S_n}$:

$$\mathbb{E}(Z_T) = \mathbb{E}\left(\left(\frac{q}{p}\right)^{S_T}\right) = \mathbb{E}(Z_0) = 1$$

which is equivalent to

$$\mathbb{P}(S_T = -a) \left(\frac{q}{p}\right)^{-a} + \mathbb{P}(S_T = b) \left(\frac{q}{p}\right)^b = 1.$$

Together with the normalization condition $\mathbb{P}(S_T = -a) + \mathbb{P}(S_T = b) = 1$, we can solve for the probabilities:

$$\mathbb{P}(S_T = -a) = \frac{1 - (q/p)^b}{(q/p)^{-a} - (q/p)^b}, \quad \mathbb{P}(S_T = b) = \frac{(q/p)^{-a} - 1}{(q/p)^{-a} - (q/p)^b}.$$

(c) By definition, the expectation of S_T is given by

$$\mathbb{E}(S_T) = -a\mathbb{P}(S_T = -a) + b\mathbb{P}(S_T = b) = -a \frac{1 - (q/p)^b}{(q/p)^{-a} - (q/p)^b} + b \frac{(q/p)^{-a} - 1}{(q/p)^{-a} - (q/p)^b}.$$

Apply optional stopping theorem on $X_n = S_n - n(p-q)$ (one can prove that $X_{n \wedge T}$ is uniformly integrable):

$$\mathbb{E}(X_T) = \mathbb{E}(S_T - T(p-q)) = \mathbb{E}(S_T) - \mathbb{E}(T)(p-q) = \mathbb{E}(X_0) = 0.$$

Hence,

$$\mathbb{E}(T) = \frac{\mathbb{E}(S_T)}{p-q} = \frac{b((q/p)^{-a} - 1) + a(1 - (q/p)^b)}{(q/p)^{-a} - (q/p)^b} \cdot \frac{1}{p-q}.$$